

AAROHAN 2026

Ascent of Knowledge

IX

Sample
Paper

Name

Roll No.

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BITS Pilani
Pilani Campus

APOGEE 2026

Instructions: -

- Read the passage carefully, it contains all the necessary details for solving the questions that follow it.
- The questions will be based on Physics, Chemistry, Mathematics and Analytical Skills.
- The questions can test your understanding of one or more of the above subjects at once.
- Questions marked with an asterisk(*) **may** have more than one option as the correct answer(s). You must select all correct option(s) to earn marks.
- **All other questions have only one correct alternative.** The marking scheme is
 - +3 -> Correct Answer
 - 0-> Unattempted Question
 - -1 -> Incorrect Answer
- For **multiple correct answer** type questions the marking is scheme is
 - +4 -> All correct options selected
 - 0 -> All other cases
- If an assumption is to be made, it will be mentioned in the question itself, along with all other necessary data/information required. Try to crack with the information that has been given- we test your **thinking process** not rote knowledge.
- Before you start this exam, **calm down** and remind yourself that **you can crack this! Keep moving forward, do not get stuck on one question.**
- **Best of Luck!!**



Flying Bike: A Basic Understanding

A hover bike is a combination between a motorcycle and a helicopter. It looks like a simple bike. This ducted-fan type bike is easy to operate and can be applied to various purposes because it does not need a runaway and is capable of hovering from any terrain.



Hover Bike is a compact aerial vehicle with a ducted fan configuration which can be able to survey and spy the surroundings and also a simple design is used for commercial purposes like a motor bike.

This is a simple construction of an overlapped propeller quadcopter Hover Bike. It consists of a coaxial type of propeller mounted on a high speed brushless electric motor and powered by rechargeable Li-Po fuel cells. It is designed with a ducted fan so that the slip of air is less. The main goal of the practical design is to develop a lightweight frame, mount an engine with propellers on it, and implement a drive system to rotate the propellers and achieve lift.

The main principle behind the working of this new breed of vehicle is “3rd LAW OF MOTION”. A Hover Bike is a type of aircraft that uses rotating, or spinning, wings called blades to fly. In order to fly, an object must have “lift,” a force moving it upward. Wings are curved on top and flatter on the bottom. This shape is called an airfoil. That shape makes air flow over the top faster than under the bottom. As a result, there is less air pressure on top of the wing. The Hover Bike is similar to a quad copter. Control of the hover bike is accomplished by turning the rotors in a particular direction. The balance of flight can be achieved as like helicopters. In Hover Bike, it is accomplished by varying the speed of the four motors relative to each other.

The rotor blades of a Hover Bike are powered by batteries. As the propeller starts to spin around, a high-pressure region is generated below the blade and a relatively low pressure region is generated above the blade. This produces an upward force called lift on the rotor blades.

The airfoils on the rotor blades generate lift that overcomes the weight of the Hover Bike, pushing it up into the air. If the lift is greater than the weight, the Hover Bike climbs; if it's less than the weight, the Hover Bike falls. When the lift and the weight are exactly equal, the Hover Bike hovers in mid-air.

Questions:-

Q1. Altitude vs. Lift force if plotted on co-ordinate axes is a straight line. Using the data provided, choose the correct value that can replace the question mark in the table given below.

- A) 181.22
- B) 179.88
- C) 165.42
- D) 180.1

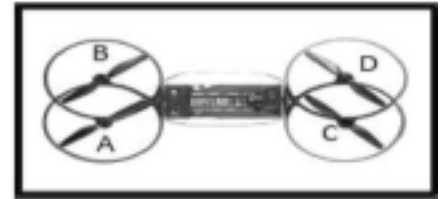
Height (feet)	Altitude (m)	Lift (L1) Kg m/s ² or N
0	0	167.52
1000	305	178.94
2000	610	179.80
3000	914	179.24
4000	1219	179.67
5000	1524	?

Q2. As can be seen from the paragraph, the flying bike has blades. The following table describes a combination of various parameters for blades. Choose the option that fits best for designing the most efficient flying bike.

Option	No. of blades	Size of blade	Material of blade
A)	High	Small	Low weight
B)	High	Large	High weight
C)	Low	Small	Low weight
D)	Low	Does not depend	High weight

Q3. The top view of the front side of the hoverboard is shown. Now, to move backward, in which component should the speed be increased and in which of them should it be decreased?

- A) Increase on 'A' and 'B' and decrease on 'C' and 'D'.
- B) Increase on 'C' and 'D' and decrease on 'A' and 'B'.
- C) Increase in all.
- D) Increase on 'A' and 'C' and decrease on 'B' and 'D'.



Q4. As can be seen in the design of the bike, the blades of the hover bike are overlapping. Choose, from the following, the most appropriate reason why.

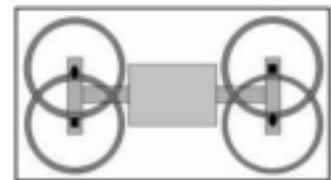
- A) For reducing the surface area.
- B) For compact size.
- C) For better design.
- D) For helping maintain balance in air

Q5. If we decrease the speed of propeller 'B' and increase that of 'C', what would happen to the hover bike?

- A) It will go down
- B) It will go up.
- C) It will go right.
- D) It will go left.

Q6. For the given figure, what value of radius 'r' of the blades given yields the most effective flying bike?

- A) 10 cm
- B) 11 cm
- C) 8 cm
- D) 9cm



**Q7. If you want to design a hover bike, you need to have an idea of the density of air as altitude changes, so that you can effectively determine the lift.

- A) Decreases significantly for each meter.
- B) Increases significantly for each meter.
- C) Decreases slightly for each meter.
- D) Increases slightly for each meter.



**Q8. For a front wing, in the left, which can bear a maximum weight of 180 kg at speed 60 km/hr, coefficient of Lift is defined as- The lift coefficient (C_L , C_N or C_Z) is a dimensionless coefficient that relates the lift generated by a lifting body to the fluid density around the body, the fluid velocity and an associated reference area. Now, as the flying bike goes up in the air, how do you think altitude varies with the coefficient of lift?

- A) It increases linearly
- B) It increases and then decreases
- C) It decreases, starting from maximum
- D) It remains constant



TRANSPORTATION

Rotational dynamics is a branch of physics that deals with the motion of objects rotating around an axis. Key terms in this field include angular velocity (ω), which describes the rate of rotation; angular acceleration (α), which represents the change in angular velocity over time; moment of inertia (I), a measure of an object's resistance to rotational acceleration; and torque (τ), the rotational equivalent of force. The fundamental equation in rotational dynamics is $\tau = I\alpha$, analogous to Newton's Second Law for linear motion ($F = ma$). Another important concept is angular momentum (L), given by $L = I\omega$. The work-energy theorem for rotational motion states that the change in rotational kinetic energy ($\Delta KE = \frac{1}{2}I\omega^2$) equals the work done by the net torque. These principles allow us to analyse complex rotational systems, from spinning tops to planetary motion.

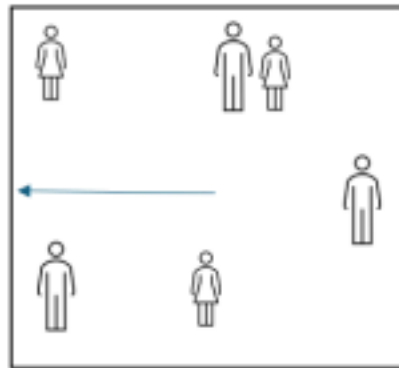
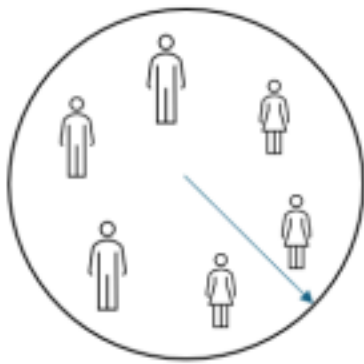
Cyclo Cable is an assistive device that is used to assist in climbing inclines while going up or down a slope. It provides a constant force to help the rider climb the incline overcoming all the external forces like friction along the way.



In order to capture the scenic views of the city while travelling, civil transportation roadway chambers are installed of different shapes and sizes and they rotate on their axis so that people can get a view while they sit in these chambers.

Questions:-

Q1. If 2 ropeway chambers, one cylindrical and the other cubic are found to have malfunctioned and they start rotating on their axes with similar but dangerously high angular velocities, which of the three people will experience more cumulative centrifugal force?



- A. People in the Cylindrical Chamber
- B. People in the Cubic Chamber
- C. Both will experience the same Force
- D. Cannot be determined

Q2. A circular chamber at an amusement park rotates at 10 radians per second around a vertical axis. A rider stands 4 metres from the centre and must press an emergency stop button located on the outer wall, 5 metres away. As the chamber rotates, the rider experiences certain forces and challenges when trying to hit the button by throwing a small object.

Which of the following statements best describes the forces acting on the rider and the correct action they should take to hit the button?

- A. The rider feels an outward force due to the rotation and should aim directly outward from the centre to hit the stop button.
- B. The rider does not feel any force inside the chamber, and they should aim directly toward the stop button to hit it.
- C. The rider feels an outward force due to the rotation and must aim slightly ahead in the direction of the chamber's rotation to ensure the object reaches the stop button.
- D. The rider feels a tangential force along the direction of their motion, so they should aim slightly behind the direction of rotation to hit the stop button accurately.

Q3. Now that the stop button has been pressed, the chamber rotationally decelerates according to the relationship $\alpha = \alpha_0 \sqrt{t}$ where $\alpha_0 = 3 \text{ rad/s}$. How long will it take for the chamber to stop?

- A. 4 sec
- B. 0.8667 sec
- C. 2 sec
- D. 1.414 sec

Q4. While the bicycle moves up the incline, in what direction does the frictional force act?

- A. Front wheel – Backwards, Rear Wheel – Forwards
- B. Front wheel – Forwards, Rear Wheel – Forwards
- C. Front wheel – Forwards, Rear Wheel – Backwards
- D. Front wheel – Backwards, Rear Wheel – Backwards

Q5. Normally while pedalling the bicycle, in what direction does the frictional force act?

- A. Front wheel – Backwards, Rear Wheel – Forwards
- B. Front wheel – Forwards, Rear Wheel – Forwards
- C. Front wheel – Forwards, Rear Wheel – Backwards
- D. Front wheel – Backwards, Rear Wheel – Backwards

Q6. Assume a passenger of mass 40 kgs travelling up a slope of height 50m and incline angle 30 degrees. The coefficient of Friction is given by the $f_s = f_k = 0.1x$ where x is the distance travelled up the incline. The Cyclo Cable assembly supplies a constant force of 24.1 N, what will be the passenger's speed when he/she is halfway along the way?

- A. 7.2 km/hr
- B. 21.6 km/hr
- C. 14.4 km/hr
- D. 20.0 km/hr